

Lecture 3: Model-Free Reinforcement Learning: Monte-Carlo, SARSA

Anton Plaksin

Markov Property

$$\mathbb{P}[S_{t+1}|S_t, A_t] = \mathbb{P}[S_{t+1}|S_0, A_0, S_1, A_1 \dots, S_t, A_t]$$

$$\mathbb{P}[R_t|S_t, A_t] = \mathbb{P}[R_t|S_0, A_0, S_1, A_1 \dots, S_t, A_t] = 1$$

Markov Decision Process

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Markov Decision Process $\langle \mathcal{S}, \mathcal{S}_F, \mathcal{A}, \mathcal{P}, \mathcal{P}_0, \mathcal{R}, \gamma \rangle$

- ▶ $\mathcal{S}$ is a **finite** ($|\mathcal{S}| = n$) state space
- ▶ $\mathcal{S}_F$ is a set of final states
- ▶ $\mathcal{A}$ is a **finite** ($|\mathcal{A}| = m$) action space
- ▶ $\mathcal{P}$ is an **unknown** transition probability function

$$\mathcal{P}(s'|s, a) = \mathbb{P}[S_{t+1} = s'|S_t = s, A_t = a]$$

- ▶ $\mathcal{P}_0$ is an **unknown** initial state probability function
- ▶ $\mathcal{R}$ is an **unknown** reward function

$$\mathcal{R}(s, a) = R_t \quad \Leftrightarrow \quad \mathbb{P}[R_t|S_t = s, A_t = a] = 1$$

- ▶ $\gamma \in [0, 1]$ is a discount coefficient

Frozen Lake World (OpenAI GYM)



Agent

(1) Action (right, left, up down)



(2) state, reward



S	F	F	F
F	H	F	H
F	F	F	H
H	F	F	G

Environment

Policy Iteration

Let π^0 and $L, K \in \mathbb{N}$.

For each $k \in \overline{0, K}$, do

► **Policy evaluation.** Iterative Policy Evaluation:

$$v^{l+1}(s) = \sum_a \pi(a|s) \left(\mathcal{R}(s, a) + \gamma \sum_{s'} \mathcal{P}(s'|s, a) v^l(s') \right), \quad l \in \overline{0, L-1}.$$

for all $s \in \mathcal{S}, \quad l \in \overline{0, L-1}$.

Define $q^L(s, a)$ by $v^L(s)$

► **Policy improvement.** Greedy Policy Improvement:

$$\pi^{k+1}(a|s) = \begin{cases} 1, & \text{if } a \in \operatorname{argmax}_{a' \in \mathcal{A}} q^L(s, a') \\ 0, & \text{otherwise} \end{cases}$$

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Main RL Algorithm Steps

- **Policy Evaluation**

Estimate v_π or q_π

- **Policy Improvement**

Update the policy: $\pi_{\text{new}} > \pi_{\text{old}}$

Q-Function along Trajectory. Deterministic Case

- Set π

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to find $q_\pi(s, a) = \mathbb{E}_\pi[G \mid S_0 = s, A_0 = a]$

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Monte-Carlo Policy Evaluation

Let π be fixed. Set $Q(s, a) = 0$.

For each k , do

- ▶ According to π , get a trajectory $\tau = (S_0, A_0, \dots, S_T)$ and rewards $(R_0, \dots, R_{T-1})$. Define $(G_0, \dots, G_{T-1})$.

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- ▶ For each $t \in \overline{0, T-1}$, update Q and N :

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha(G_t - Q(S_t, A_t))$$

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$$Q(s, a) \approx q_\pi(s, a)$$

Using Bellman Equation

Bellman Expectation Equation for q_π

$$q_\pi(s, a) = \mathcal{R}(s, a) + \gamma \sum_{s'} \mathcal{P}(s'|s, a) \sum_{a'} \pi(a'|s') q_\pi(s', a')$$

$\Downarrow$

$$q_\pi(s, a) = \mathbb{E}_\pi[R_t + \gamma q_\pi(S_{t+1}, A_{t+1}) \mid S_t = s, A_t = a]$$

$\Downarrow$

Temporal-Difference

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha(R_t + \gamma Q(S_{t+1}, A_{t+1}) - Q(S_t, A_t))$$

Temporal-Difference Policy Evaluation

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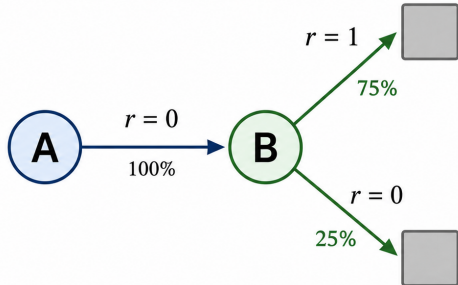
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Comparison of MC and TD Policy Evaluation

Two states A , B ; no discounting; 8 episodes of experience

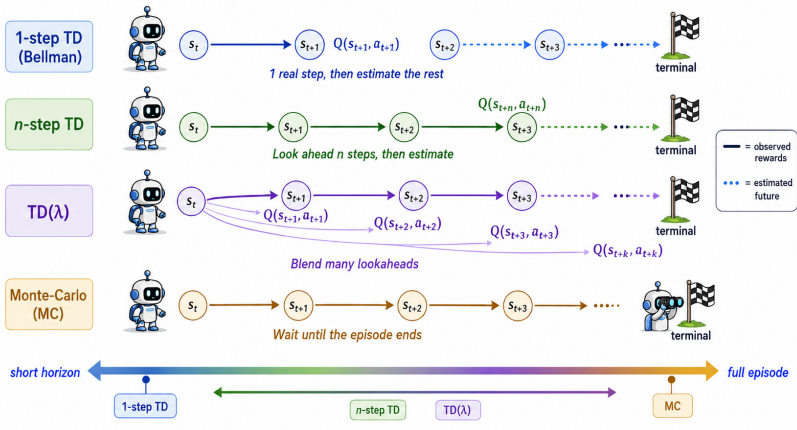
8 episodes of experience

- 1 $A, 0, B, 0$
- 2 $B, 1$
- 3 $B, 1$
- 4 $B, 1$
- 5 $B, 1$
- 6 $B, 1$
- 7 $B, 1$
- 8 $B, 0$



? What is $V(A)$, $V(B)$?

More Policy Evaluation Ways



Q-function Evaluations

- ▶ Monte-Carlo (MC).

$$Q(S_t, A_t) \approx G_t, \quad G_t = \sum_{k=t}^{T-1} \gamma^{k-t} R_k$$

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- ▶ n -step TD. It is "Bellman" Evaluation for $n = 1$.

$$Q(S_t, A_t) \approx G_t^{(n)}, \quad G_t^{(n)} = \sum_{k=0}^{n-1} \gamma^k R_{t+k} + \gamma^n Q(S_{t+n}, A_{t+n})$$

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- ▶ TD(λ).

$$Q(S_t, A_t) \approx G_t^{(\lambda)}, \quad G_t^{(\lambda)} = (1 - \lambda) \sum_{n=1}^{\infty} \lambda^{n-1} G_t^{(n)}$$

TD(0) = 1-step TD; TD($\lambda \rightarrow 1$) $\rightarrow$ MC.

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V-function Evaluations

By replacing $Q(S_t, A_t)$ with $V(S_t)$ we obtain exactly the same formulas for evaluating value function

Will This Work?

Let π^0 be initialized and $K > 0$.

For each $k \in \overline{1, K}$, do

- **Policy evaluation.** Any Policy Evaluation.

Obtain $Q^k(s, a) \approx q_{\pi^k}(s, a)$

- **Policy improvement.** Greedy Policy Improvement:

$$\pi^{k+1}(a|s) = \begin{cases} 1, & \text{if } a \in \operatorname{argmax}_{a' \in \mathcal{A}} Q^k(s, a') \\ 0, & \text{otherwise} \end{cases}$$

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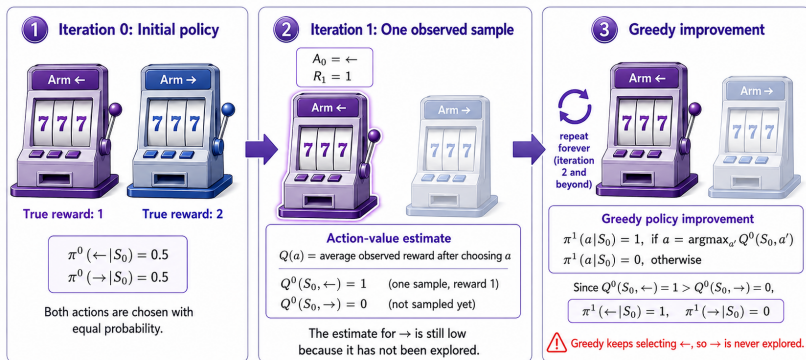
Obtain $Q^k(s, a) \approx q_{\pi^k}(s, a)$

- **Policy improvement.** Greedy Policy Improvement:

$$\pi^{k+1}(a|s) = \begin{cases} 1, & \text{if } a \in \operatorname{argmax}_{a' \in \mathcal{A}} Q^k(s, a') \\ 0, & \text{otherwise} \end{cases}$$

NO.

Why Greedy Improvement Can Fail



Estimated best action $\neq$ truly best action

After a single sample, greedy improvement can stop exploration too early and lock in a suboptimal action.

Exploration vs Exploitation

Problem

Greedy policy improvement uses only current estimates:

$$\pi(a|s) = \begin{cases} 1, & \text{if } a \in \operatorname{argmax}_{a' \in \mathcal{A}} Q(s, a') \\ 0, & \text{otherwise} \end{cases}$$

- **Exploitation:** choose actions that look best according to current Q .
- **Exploration:** choose actions to collect more information.

Trade-off

Too much exploitation may miss better actions. Too much exploration slows down learning and reduces reward during training.

Goal

Choose good actions often, but still try uncertain or non-greedy actions.

► ϵ -greedy.

$$\pi(a|s) = \begin{cases} 1 - \epsilon + \epsilon/m, & \text{if } a \in \operatorname{argmax}_{a' \in \mathcal{A}} Q(s, a'), \\ \epsilon/m, & \text{otherwise} \end{cases}$$

► Softmax / Boltzmann exploration.

$$\pi(a|s) = \frac{\exp(\tau Q(s, a))}{\sum_{a'} \exp(\tau Q(s, a'))}.$$

► Upper Confidence Bound.

$$A_k = \operatorname{argmax}_{a \in \mathcal{A}} \left(Q(s, a) + c \sqrt{\frac{\ln k}{N_k(s, a)}} \right).$$

Main RL Algorithm Steps

- ▶ **Policy Evaluation**

Estimate v_π or q_π

- ▶ **Policy Improvement**

Update the policy: $\pi_{\text{new}} > \pi_{\text{old}}$

$$\pi = \varepsilon\text{-greedy}(Q)$$

$$\pi(a|s) = \begin{cases} 1 - \varepsilon + \varepsilon/m, & \text{if } a \in \operatorname{argmax}_{a' \in \mathcal{A}} Q(s, a'), \\ \varepsilon/m, & \text{otherwise} \end{cases}$$

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Policy Improvement Theorem

Let $Q(s, a)$ be defined. Let $\pi = \varepsilon\text{-greedy}(Q)$ and $\pi' = \varepsilon\text{-greedy}(q_\pi)$.

Then

$$\pi' \geq \pi, \quad (v_{\pi'}(s) \geq v_\pi(s), \quad \forall s)$$

SARSA Algorithm

Let $Q(s, a) = 0$, $K > 0$, and $\varepsilon = 1$.

For each k , do

During trajectory

- From the state S_t , acting $A_t \sim \pi(\cdot|S_t)$,
where $\pi = \varepsilon$ -greedy(Q), get R_t , go to the next state S_{t+1} ,
and act $A_{t+1} \sim \pi(\cdot|S_{t+1})$
- According to $(S_t, A_t, R_t, S_{t+1}, A_{t+1})$, update Q :

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha_k(R_t + \gamma Q(S_{t+1}, A_{t+1}) - Q(S_t, A_t))$$

Put $\varepsilon = 1/k$

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Theorem

Let each (s, a) be visited infinitely often and $\sum_k \alpha_k = \infty$, $\sum_k \alpha_k^2 < \infty$.
Then $\pi \rightarrow \pi_*$ as $k \rightarrow \infty$

QUESTIONS?